

SK MD SHAHID

DATA SCIENTIST

Bhubaneswar, Odisha

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SUMMARY

Data Scientist and AI Engineer with a strong background in Physics and Mathematics, holding a Master's degree in Data Science & AI. Experienced in AI research, machine learning, and NLP, with a proven track record of delivering innovative solutions in healthcare and startup environments. Seeking an entry-level position to contribute technical expertise and analytical skills to cutting-edge AI projects.

WORK

EXPERIENCE

Lead AI/ML Engineer | Brillio

June 2025-Present

- **Enterprise Code Conversion Accelerators** : Conceptualized and built enterprise-level AI accelerators designed to automate large-scale code and database migration pipelines.
- **Agentic ETL & Data Discovery** : Developed multi-agent workflows for automated data discovery, structural ETL mapping, and seamless target-system schema integration.
- **Automated Code Generation** : Leveraged advanced LLM routing to dynamically generate optimized source code tailored precisely to target system architecture and constraints.

Data Scientist/GenAI Engineer | Shenzyn

December 2024-June 2025

- **Threat Hunting Assistant** : Automated threat detection and reporting in AWS Security Data Fabric (Petabyte-scale) OCSF mapped security logs aid the threat hunting team of fortune-50 corp.
 - * Orchestrated workflows using **LangGraph** and stateful execution with integration across AWS services like Athena, S3, Lambda, and EventBridge.
 - * Utilized **AWS Bedrock** Client for Claude LLM and FastAPI backend service, with DynamoDB for database management.
 - * Improved detection speed and accuracy, enabling the team to process vast amounts of security data effectively.

Trainee Software Engineer | Nitor Infotech

May 2024-August 2024

- **GE- CoPilot** :Automated workflows for a biotech grant writing company, reducing manual processing time from **7 hours to approximately 5 minutes**, improving operational efficiency by over **98%**, and enhancing data security.
 - * Designed a customized in-house co-Pilot using **LangChain LLM Agents, FastAPI, AWS EC2, and Docker**.
 - * Integrated external databases like **PubMed and NIH** and deployed with **MongoDB and React.js** frontend.
 - * Application currently utilized by 40-50 researchers, accelerating research productivity.
- Developed PoCs integrating **AI agents and knowledge graphs** to enhance healthcare analytics and survival analysis.
 - * Built solutions using **LangChain, LangGraph, Neo4j, Gremlin API, and advanced chain of thought prompting techniques**.

AI Researcher | OdiaGenerative AI

May 2023- May 2024

- Co-developing **FEDCOM (Framework for Easy Deployment of Compressed and Optimized Models)**, a collaboration between **Odia Generative AI** and the **Norwegian BioAI Lab**, aimed at simplifying deep learning model compression and deployment in resource-constrained environments.
- Developed **OdiaLLAMA, a LLama-based Odia language model**, and **Acharya, an RAG-based Hindi tutor**, advancing low-resource language AI.
 - * Utilized **LoRA fine-tuning, Axolotl, and RAG techniques for training and deployment on AWS, Streamlit, and Hugging Face Spaces**.

Ongoing Research work | OdiaGenerative AI

May 2023- May 2024

- Leading research initiatives focused on low-resource Indic languages, spanning OCR optimization, Vision-Language Models (VLMs), and foundational LLM pre-training.
- Developing robust Evaluation Benchmarks and curating high-quality pre-training datasets to standardize multilingual performance.
- Implementing state-of-the-art LLM quantization techniques to facilitate efficient, resource-constrained edge deployment.
- Co-developing **FEDCOM (Framework for Easy Deployment of Compressed and Optimized Models)**, a collaboration between **Odia Generative AI** and the **Norwegian BioAI Lab**.

- Developed **OdiaLLAMA**, a LLama-based Odia language model, and **Acharya**, an **RAG-based Hindi tutor**, advancing low-resource language AI.
 - * Utilized **LoRA fine-tuning**, **Axolotl**, and **RAG techniques** for training and deployment on **AWS**, **Streamlit**, and **Hugging Face Spaces**.

EDUCATION	Master of Science (Data Science & AI) Silicon University, Bhubaneswar 2022-2024 Specialized in machine learning, deep learning, natural language processing, big data analytics, time series, bioinformatics.
	Bachelor of Science (Physics & Mathematics) Berhampur University, Odisha 2018-2021 Built a strong foundation in physics and mathematics, focusing on statistical mechanics, quantum physics, and advanced calculus.
SKILLS	Programming languages : Python (Proficient), R (Intermediate) Data Science & Machine Learning : Machine Learning Algorithms (Regression, Classification, Clustering), Deep Learning (CNNs, RNNs, Transformers), Natural Language Processing (LLMs, Text Generation, Language Translation), Time Series Analysis, Generative AI Techniques, Document AI, OCR, LLM Pretraining AI Frameworks & Libraries : TensorFlow, PyTorch, scikit-learn, Keras, Langchain, Langgraph, Knowledge Graphs, Multi-Agent Framework, Advanced RAG, LLM Evaluation, Llama Factory (LLM Finetuning), NLTK, SpaCy, Model Context Protocol (MCP) Data Analysis & Visualization : Numpy, Pandas, Seaborn, Matplotlib, Plotly, Tableau Databases : SQL (MySQL, PostgreSQL), NoSQL (MongoDB). Cloud & DevOps Tools : AWS (EC2, S3, Lambda), Docker, GCP, VertexAI, Azure, Azure AI Foundry, AWS Bedrock, Microsoft Fabric Software Development : Backend Development (FastAPI, Flask), API Development and Integration, Version Control (Git, GitHub) Mathematical Foundations : Linear Algebra, Statistics & Probability, Optimization Techniques, Graph Theory, Advanced Calculus Theoretical Knowledge : Big Data, Data Security & Privacy, Bioinformatics
ACTIVITIES & ACHIEVEMENTS	Publications & Presentations • Poster : "LLM Fine-Tuning for Low Resource Language : A Case Study on Odia" - Presented at Finnish AI Day 2023, November 2023. 🔗 • Paper : "OdiaGenAI's Participation at WAT2023" - Published in the conference proceedings of MT Summit 2023. 🔗 • Workshop-WAT 2023 : Kyoto University and NICT, hosted by MT Summit 2023, Macau SAR China, September 2023. 🔗 • Conference-MT Summit 2023 : Macau SAR China. 🔗 • OdiagenAI's Work Presentation : Odia's in AI ML Global Conference, October 2023. 🔗 • Generative AI and LLM Workshop : Conducted by OdiagenAI, November 2023. 🔗 Media Featured • "OdiaLLama by OdiagenAI" featured in Analytics India Magazine, December 2023. 🔗 • "OdiaLLama by OdiagenAI" featured in Meta Community Stories, December 2023. 🔗 Hackathons & Competitions Secured top 5% in ML hackathon by Analytics Vidya .